



Interdisciplinary Deep Learning On Graphs

Course Project Review

Node-to-Self

Yash Verma

PES1UG23AM910 | Section D

Interdisciplinary Deep Learning on Graphs

PromptGFM-Bio: Prompt-Conditioned Graph Foundation Model for Rare-Disease Gene Mapping



Rare diseases affect over 300 million people worldwide, yet most have very few known causal genes. Traditional gene discovery methods struggle with the long-tail distribution where thousands of diseases have extremely limited labeled data.

PromptGFM-Bio addresses this by combining a Graph Neural Network (GraphSAGE) operating on a heterogeneous biomedical knowledge graph with a BioBERT-based prompt encoder that injects disease-specific context via FiLM conditioning. Given a disease description, the model ranks all candidate genes by their predicted association strength.

The same graph backbone produces different predictions depending on the disease prompt, enabling adaptive reasoning across diverse rare disease queries from a single trained model.

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What Makes This Project Unique?



Prompt-Conditioned Graph Learning: Disease descriptions encoded by BioBERT serve as active conditioning signals via FiLM modulation, not just static feature concatenation.

Task-Specific Graph Reasoning: The same biological graph produces different gene rankings depending on the disease prompt, enabling adaptive reasoning per query.

Heterogeneous Biomedical Backbone: Built on a large-scale heterogeneous graph with 5,363 genes, 16,841 diseases, 11,794 phenotypes, and 1.17M curated gene-disease edges.

Multiple Conditioning Strategies: Supports FiLM, cross-attention, and hybrid fusion strategies for ablation comparison.

Rare-Disease Focus: Designed for the long-tail problem with support for rarity-stratified evaluation and few-shot gene discovery splits.

Combined Loss Objective: Jointly optimizes BCE classification loss and margin ranking loss for better gene ranking quality.

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Datasets and Graph Structure



Data Sources:

- BioGRID and STRING: Protein-protein interaction (PPI) networks for gene-gene edges
- HPO (Human Phenotype Ontology): Gene-to-phenotype and disease-phenotype annotations
- Orphadata: Gold-standard rare disease gene associations for validation

Processed Graph (PyTorch Geometric HeteroData):

- Node types: 5,363 genes | 16,841 diseases | 11,794 phenotypes
- Supervised edges: 1,170,143 gene-disease edges (HPO phenotype bridge, IDF-weighted)
- Splits: 936K train | 117K val | 117K test (80/10/10) with 5 negative samples per positive

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Project Progress So far

Completed:

- ✓ Full end-to-end pipeline: data download, preprocessing, training, and evaluation scripts
- ✓ PromptGFM model: GraphSAGE backbone + BioBERT prompt encoder + FiLM conditioning + MLP predictor
- ✓ Trained 100 epochs on RTX 4090 with AMP, cosine LR schedule, early stopping at epoch 46
- ✓ Baseline results: AUROC 0.813 | AUPR 0.462 | Precision@10 = 1.0 | NDCG@100 = 0.929

Identified Improvements (Next Steps):

- Activate GNN message passing by incorporating PPI gene-gene edges from BioGRID/STRING
- Replace disease IDs with natural-language prompts (name + phenotype descriptions) for BioBERT
- Implement per-disease ranking evaluation and run ablation comparisons (GNN-only vs. PromptGFM)



THANK YOU

Yash Verma

Department of Computer Science and Engineering in AI & ML

PES1UG23AM910